PSEL4 compiles each H2 circuit four ways and keeps the cheapest

Over 217 benchmark circuits PSEL4 needs 15.7 % less HQC than pytket-quantinuum's default level 2. It costs more than level 2 on no circuit and never more than level 3.

Most of the saving comes from choosing level 3 where it is cheaper. On the H2-2E emulator the cheaper compile of cdkm_adder_10 returns the correct sum less often. [O] The finals goal is a selector that also learns fidelity.

Team Learned ZX Rewriting

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The default compile of pytket-quantinuum is not the cheapest on H2

A run on Quantinuum H2 is billed in HQC.

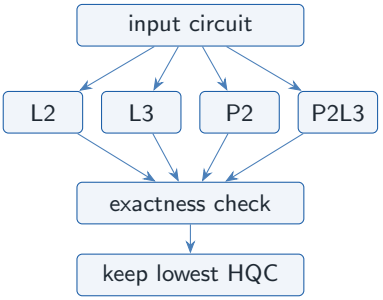
$$\mathrm{HQC} = 5 + C\left(N_{1q} + 10\,N_{2q} + 5\,N_{\mathrm{SPAM}}\right)/5000$$

C is the shot count. N_{1q} counts PhasedX gates. N_{2q} counts ZZPhase and ZZMax gates. N_{SPAM} counts initialisations, measurements and resets. Rz gates are free.

- pytket-quantinuum compiles at optimisation level 2 by default. Level 3 adds GreedyPauliSimp, a two-qubit resynthesis.
- Always using level 3 saves 13.9 % HQC over 217 circuits, yet it costs more than level 2 on 45 of them.
- ZX rewriting (PyZX teleport_reduce) before the Quantinuum pass lowers the cost on some circuits and raises it on others.
- Qiskit at optimisation level 3, rebased to H2, costs 25.24 % more than level 2 on B1 and 25.16 % more on family C.
- **[O]** A team with a fixed HQC budget cannot tell in advance which compile is cheapest for a given circuit.

PSEL4 keeps the lowest HQC of four checked compilations

- **L2 (P0).** pytket-quantinuum 0.59.3 at level 2, the default.
- **L3.** The same pass at optimisation level 3.
- **P2 and P2L3.** PyZX teleport_reduce, then level 2 or level 3.
- **Exactness check.** Full unitary up to 8 qubits, all-zero input state from 9 to 20 qubits. Failed candidates are dropped.
- **Guarantee.** Level 2 and level 3 are always candidates. PSEL4 never costs more HQC than either where both pass the check. This holds by construction.



`pipelines/pass_select.py --psel4`

Assessment [O]. PSEL4 is pytket's PassSelector pattern with HQC as the metric. The contribution is the measured cost ledger, the correctness evidence and the emulator tests.

Finding 1. PSEL4 needs 15.7 % less HQC than always using level 2

		Total HQC			Saved vs always L2 %		
Family	Circuits	Always L2	Always L3	PSEL4	Always L3	PSEL4	L3 loses on
A, Trotter	180	16,422.6	16,231.2	15,949.0	1.2	2.9	35
B1, Clifford+T	15	15,516.0	13,965.8	13,863.4	10.0	10.7	2
C, arithmetic and oracles	20	24,535.0	18,853.2	18,193.2	23.2	25.8	8
Grover use case	2	2,363.6	1,582.0	1,575.0	33.1	33.4	0
Pooled	**217**	58,837.2	50,632.2	49,580.6	13.9	**15.7**	**45**

HQC at 1000 shots. PSEL4 costs more than level 2 on 0 of 217 circuits. Source `value/policy_compare.md`.

- Always using level 3 loses 372.0 HQC on 45 circuits. The largest loss is 12.95 % on gf2_pow_4_mult.
- PSEL4 saves 1,051.6 HQC more than always using level 3, or 1.79 % of the level 2 total.
- PSEL4 runs four compilations. Its total wall time is 5.08 times that of level 2 over 215 timed circuits. It saves 3.03 HQC per extra second.

Assessment [O]. For a cost-bound H2 user PSEL4 is a drop-in that cuts HQC by 15.7 % on this benchmark at about 5 times the compile time of level 2.

Most level 3 and PSEL4 outputs are proven equal to their input

Family C output	Proven	Not proven	Saved vs L2 (proven)
Level 3	18 of 20	hrs_mult_5, rgqft_mult_6	13.33 %
PSEL4	17 of 20	hrs_mult_5, rgqft_mult_6, draper_adder_28	10.22 %

Family C has 20 to 56 qubits. A PyZX miter reduction proves equality of the full unitary up to global phase. The level 3 miters of the two unproven circuits timed out after 900 s. Sources `results/family_C_equivalence_L3.md`, `value/psel4_zx_only.md`.

- hrs_mult_5 has off-grid angles. Its PSEL4 output passes a check on 64 random basis inputs only. The PSEL4 miters of rgqft_mult_6 and draper_adder_28 timed out after 300 s.
- On families A, B1 and B2 every compiled circuit is checked against the uncompiled reference with tolerance 1e-8. All 1260 A rows, 105 B1 rows and 100 B2 rows pass.
- Our level 2 pipelines give the same counts and HQC as the official pytket-quantinuum pass on all 445 compiled rows of families A, B1, B1x and B2. Family C was not compiled with the official pass.

Finding 2. Level 3 cuts two-qubit gates and ZX cuts phases

	Non-Clifford gates				Two-qubit gates			
Set	L2	P2	L3	P2L3	L2	P2	L3	P2L3
B1, 14 circuits	877	656	746	701	862	799	723	700
C, 15 circuits	3323	2930	2540	2535	3208	2944	2718	2708

Totals after each pipeline. Source `value/mechanism_zx_l3.md`. 6 circuits are missing (compile budget).

- GreedyPauliSimp lowers the two-qubit count in level 3 on 180 of 180 A, 13 of 14 B1 and 9 of 15 C circuits.
- On B1 teleport_reduce leaves fewer non-Clifford gates than level 3 (656 against 746). This rarely lowers HQC because a two-qubit gate costs 2 HQC at 1000 shots and a PhasedX 0.2.
- The ZX-only gain on top of the cheaper pytket level is 0.26 % on the 17 proven C circuits, 3.21 % on all 20, 0.49 % on B1 and 0 on the Trotter circuits.
- It falls as T-density rises (family C Spearman $\rho = -0.64$, $p = 0.010$).

Assessment [O]. The hypothesis that level 3 absorbs the ZX gain partly holds. It holds for two-qubit gates and fails for phase merging on B1.

Finding 3. The cheaper compile is not always the more accurate

		Two-qubit gates			Correct output		
Circuit, PSEL4 pick	Input	L2	Pick	Shots per arm	L2	Pick	z
cdkm_adder_10, L3	a = 692, b = 435	141	118	400	0.8425	0.8325	-0.38
	a = 502, b = 24	141	118	400	0.8525	0.8000	-1.96
	a = 710, b = 328	141	118	400	0.9125	0.8475	-2.84
	pooled, primary			1200	**0.8692**	**0.8267**	$\mathbf{-2.90}$
modadd_2n_10, P2L3	pooled, primary	147	142	1200	0.8417	0.8383	-0.22
vbe_adder_7, L3	a = 28, b = 111	142	79	1000	0.883	0.901	1.30

H2-2E noisy emulator. Correct output is the share of shots with the correct sum. Sources `results/h22e_multi.md`, `results/h22e_famC_nz.md`.

- Inputs, picks and the primary metric were pre-registered in files committed before any job ran.
- On cdkm_adder_10 the level 3 output has fewer two-qubit gates (118 against 141) and two-qubit depth 65 against 122, yet returns the correct sum less often.
- modadd_2n_10 shows no difference. vbe_adder_7 favours level 3, not resolved at 1000 shots.

Assessment [O]. On the H2-2E emulator the cheapest compile is not always the most accurate. For accuracy-bound work the fidelity should be checked first.

The level 2 and level 3 candidates guard against costly blind rewriting

- Blind teleport_reduce before level 2 costs 6.27 % more HQC on 180 Trotter circuits and 22.1 % more on 3 QFT adders.
- On draper_adder_16 it costs 1,385.0 HQC against 906.8 for level 2.
- pytket's own `ZXGraphlikeOptimisation` followed by level 2 costs 59.22 % more than level 2 on 14 B1 circuits and 116.49 % more on family C.
- PSEL4 never keeps such a candidate when level 2 or level 3 is cheaper.

Earlier emulator tests of vbe_adder_3 used the all-zero input

Circuit	Pipeline	Two-qubit gates	Shots	F_H	sd
vbe_adder_3, all-zero input	L2 (P0)	56	4000	0.9520	0.0034
vbe_adder_3, all-zero input	P2	41	4000	0.9628	0.0030

H2-2E, pooled over four runs. Source `results/h22e_confirmatory.md`.

- P2 is higher by 0.0108 with $z = +2.38$ and two-sided $p = 0.017$.
- The Holm-adjusted p is 0.035 over the two repeated circuits and 0.052 over all three H2-2E circuits of that run.
- A repeats-only check gives a Holm-adjusted p of 0.030. It was chosen after the pooled result was known. grover_5 shows no gain.
- On the all-zero input the ideal output of vbe_adder_3 equals the input string.

Assessment [O]. The vbe_adder_3 gain is not established at 0.05 after correction over three circuits. The non-zero-input tests of Finding 3 replace it.

A known T-count bound limits further exact T optimisation

- Each T gate of a surface-code machine consumes one magic state. On H2 a T gate is a free Rz.
- On B1 the ZX stage removes 2,862 of 7,048 T gates of the written circuits, or 40.6 %.
- The floor is the known channel bound of Gosset et al. (2014), evaluated exactly. On 20 synthesised Haar circuits it is 0.997 of the T-count after ZX, pooled. Further exact T optimisation can gain at most 0.3 % pooled and 0.72 % on the worst instance.
- The evaluation covers 24 circuits of at most 5 qubits and no circuit of family C.

Limitations

- No user interview or survey is part of this submission.
- The ZX-only gain is small, 0.26 % on the 17 proven family C circuits.
- 2 family C level 3 outputs and 3 PSEL4 outputs are unproven.
- HQC is the formula estimate. All fidelity results come from the H2-2E emulator. Nothing has run on H2 hardware.
- The earlier vbe_adder_3 tests used the all-zero input. Their pooled Holm-adjusted p is 0.052 over three circuits.
- The T-count floor is the known Gosset et al. bound, evaluated on 24 circuits of at most 5 qubits.
- The selector follows pytket's PassSelector pattern.
- The mechanism analysis lacks 6 circuits whose level 3 or P2L3 compilation did not fit the CPU budget.

Finals plan is a fidelity-aware learned selector

- **[O]** The finals goal is a learned selector that predicts fidelity as well as HQC for each candidate compile, following Finding 3.
- A first classifier on input-circuit features predicts whether ZX before level 2 is cheaper. With a whole family held out it does not beat constant choices. Logistic regression reaches a pooled balanced accuracy of 0.433 against 0.500 for the constant choices.
- **[O]** The finals model therefore needs structural features and emulator labels.
- **[O]** The four-candidate ledger of 217 circuits and the 34 stored H2-2E jobs are its starting data.

Every result can be rerun from the code package

- `code.zip`. The source of the repository. `README.md` gives the run instructions.
- `requirements.txt`. Pins pytket 2.18.4, pytket-quantinuum 0.59.3 and pyzx 0.10.7.
- `run_tests.sh`. Runs 28 test files. The last run in `submission/build_logs/tests_summary.txt` shows rc=0 for all 28.
- `EVIDENCE.md`. Lists every emulator job with its job ID, shots, HQC estimate and stored payload. 34 H2-2E jobs used an HQC estimate of 8983.4. Of the 10,000 HQC budget 1016.6 are left.
- **Generated tables.** The value tables come from the scripts under `value/`. The emulator tables are regenerated offline from the stored payloads.
- **Independent checks.** Verification reports by agents that did not produce the results are in `results/`.